

Writer's backend briefing

Internal notes for a casual-but-accurate 2026 article. Not copy. Numbers below are from the live code and the last walk-forward (trained through 2025, board for 2026). Date of this dump: 20 Aug 2026.

What you are writing about: a 12-team full-PPR ranking model whose job is not “nail every player’s point total.” It is “rank drafted players relative to ADP / Vegas, and paint a name green or red only when the gap is large and there is a football reason.”

Live board: <https://abhinav811.github.io/NFLDraftModel/> · GitHub: [Abhinav811/NFLDraftModel](#) · Pages from `main` / `docs`.

Do not claim in the article

- That this is a betting model, or that VFP is +EV on props.
- That point MAE (~ 55 PPR) is “accurate crystal ball” — that error is normal for season-long fantasy; the claim is ranking lift vs ADP.
- That the model beats ADP every year. 2021 was a tiny loss (-0.005 Spearman); 2024 was a tie ($+0.001$). Mean lift $\sim +0.019$.
- That every ADP disagreement is a steal. Close-to-ADP names are a coin flip. Flags need ± 5 position ranks *and* ± 12 PPR *and* a confirmer.
- That Half PPR is a second trained model. It is full PPR minus $0.5 \times$ projected receptions.
- That midseason is a separate product. Same board; live team + remaining SOS move names in place. Steal/fade flags stay frozen.

1. One-sentence architecture

For each skill player (QB/RB/WR/TE) in season t , build a market number m (Vegas fantasy points if a prop exists, else an ADP-implied points curve), mix in a tenure-weighted prior (production, situation, physical, aging), then add a small gradient-boosted residual that tries to predict how last year’s m was wrong. Tune how much residual to trust (λ) on ranking correlation, not on how many names turn green.

```
 $\hat{y} = \text{market\_or\_blend} + \lambda_{\text{tenure}} \cdot \text{residual} + k \cdot \text{rank\_lift}$   
then a little extra pull toward the prior for rookies without a prop
```

Walk-forward: train only on seasons < t, score t. Never use 2026 outcomes (they don't exist). Historical *m* is ADP because there is no archive of player-level season props.

2. Pipeline (what actually runs)

1. `python3 scripts/run_pipeline.py` — download nflverse + ADP + props, build `data/processed/player_panel.parquet` (not published), walk-forward backtest, fit 2026, write rankings CSVs + methods HTML + rebuild `docs/index.html`.
2. `python3 scripts/refresh_board.py` — no retrain. Re-rank the saved 2026 CSV with live Sleeper injuries/teams and remaining SOS. Used by GitHub Action daily 14:00 UTC.
3. Push `docs/index.html` (and figures) to main for Pages.

Processed parquet is gitignored on purpose. The hosted page is a static HTML blob of ~192 ADP-drafted names.

3. Where every number comes from

Layer	Source	How it is used
Play-by-play	nflverse <code>play_by_play_YYYY.parquet</code> (CC BY 4.0)	Red zone, chunk plays, O-line index, PROE/pace, team SOS, primary passer
Season stats	nflverse <code>stats_player_reg_YYYY</code>	Lagged PPR, usage, EPA, WOPR; realized <i>y</i> for backtests
Weekly stats	nflverse <code>stats_player_week_YYYY</code>	H2 vs H1 carry and PPR deltas (lagged +1 season)
NGS	nflverse <code>ngs_{passing,rushing,receiving}.parquet</code>	QB intended air yards; RB RYOE/att; WR air-share (lagged)
Rosters / players	nflverse rosters + players	Team, position (FB/HB→RB), age, years_exp
Depth charts	nflverse <code>depth_charts</code> (weekly schema 2019–24; ESPN-style dt/pos_rank 2025–26)	Week-1 / pre-kickoff depth_rank. ADP-pool fill ~97% after the join fix

Injuries / snaps	nflverse injuries, snap_counts	Missed weeks, availability, snap %, chronic 3-yr rollup
Combine / draft	nflverse combine + draft_picks	Physical prior for young players; undrafted pick = 250
Schedule	nflverse games	This year's opponents → SOS from last year's def EPA; indoor %; implied totals/spreads
ADP	Fantasy Football Calculator API, 12-team PPR (half-PPR when toggling)	Market proxy historically; steal band; sample weights; "vs ADP"
Season props / VFP	User wide CSV data/external/season_long_props_wide.csv (FTA scrape + DraftKings often blocked)	2026 market m when ≥ 1 market posts
ECR	DynastyProcess FantasyPros archive (Aug–Sep redraft)	Tree feature ecr, ecr_minus_adp
OC map	Wikipedia team-season staff + current OC list (PFR is Cloudflare-blocked)	new_oc binary; 2026 has 31/32 teams; history is partial
Live injuries / team	Sleeper /v1/players/nfl until nflverse weekly reports exist; then nflverse injuries	Q / O / IR badges; club changes (LAR→LA etc.)

Names are normalized (strip Jr/III, accents). Teams aliased (LAR→LA, JAC→JAX, WSH→WAS, ...). Skill positions only: QB, RB, WR, TE.

4. Scoring (what “points” means)

pass yard	0.04	rush/rec yard	0.10	reception	1.00
pass TD	4.00	rush/rec TD	6.00	INT / fum	-2.00

That is full PPR. Half PPR on the board is the same $\hat{y}$ minus $\frac{1}{2} \times$ projected receptions (prop receptions if present, else last year's, else a position catch-rate $\times$ targets). Not a re-fit.

Games: 16 through 2020, 17 from 2021 on (including 2026).

5. The market layer — what “ADP” and “Vegas” actually are

5.1 Vegas Fantasy Points (VFP) — 2026 only

Posted over/under L is treated as a median. American odds $\rightarrow$ implied $p_{\text{over}} / p_{\text{under}}$, vig-stripped to a fair over probability, then a damped shift of the line:

```
p = |o|/(|o|+100) if o<0 else 100/(o+100)
p_fair = p_over / (p_over + p_under)
z = clip(  $\Phi^{-1}(p_{\text{fair}})$ , -1.5, 1.5 )
E[X] = L + 0.35 ·  $\sigma_{\text{market}}$  · z
VFP =  $\sum E[X_k]$  · scoring_k
```

σ scales (so juice cannot yank a total several σ): pass yd 450, rush yd 180, rec yd 160, rec 12, pass TD 4.5, rush TD 2.8, rec TD 2.4, INT 2.2, fum 1.2. Used as m only if $vfp_{\text{markets}} \geq 1$.

Casual angle: books are better at volume than a public model. The model is not trying to out-Vegas passing yards. It is trying to notice aging, vacated touches assigned to a *person*, TD luck vs role, cliffs, etc.

5.2 ADP-implied points (historical m , and 2026 fallback)

By position, OLS: $m_{\text{ADP}} = \beta_0 + \beta_1 \cdot \log(\text{ADP})$ fit on historical (ADP, actual PPR). Two WRs five picks apart can be 40 PPR apart. This is a *points* market, not “they’re next to each other on a cheat sheet.”

FFC 12-team PPR ADP. Sample weights in the trees: everyone 1 $\times$; $\text{ADP} \leq 150$ gets 4 $\times$; ≤ 72 gets 5.5 $\times$; steal band 18–132 gets another $\times 1.2$. The model is trained to care about draftable names, not UDFA dart throws.

6. Features a yardage prop cannot see

These feed the prior and/or the trees. Trees see ~ 100 columns plus `position_code` and `tenure_code`. Team-context columns are *pre-shrunk* so O-line/SOS cannot dominate usage: $\times 0.35$ for OL/scheme/pace, $\times 0.40$ for SOS / indoor / implied totals.

6.1 Age

Age = days from birth to Sept 1 of season t / 365.25. Peaks: QB 30, RB 26.5, WR 28, TE 28.5.

```
 $\alpha_{\text{age}} = \text{clip}(1 - \lambda_{\text{pos}} \cdot [\text{age} - \text{peak}]^2, 0.35, 1.05)$ 
```

λ is fit on players already past the peak with last-year PPR > 50. Defaults if thin: QB 0.012, RB 0.035, WR 0.018, TE 0.016. RBs fall off harder because the data said so.

6.2 Tenure buckets (prior mix only)

years_exp	Bucket	Market	Production	Situation	Physical	Aging
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0	rookie	0.22	0.08	0.40	0.30	0.00
1	sophomore	0.22	0.32	0.32	0.12	0.02
2-3	developing	0.28	0.38	0.24	0.06	0.04
4-6	prime	0.40	0.34	0.16	0.04	0.06
≥7	veteran	0.38	0.24	0.10	0.02	0.26

Rookies lean on situation + athleticism. Prime vets lean on the market + last year. Aging vets get a heavier age penalty. Trees still see raw years_exp.

```
blend = w_m·m + w_p·prod + w_s·sit + w_ph·phys + w_a·(α_age · prod)
prod = PPR_eff · α_age · (0.85 + 0.15 · availability)
PPR_eff = (PPR/G)·G_season if missed weeks ≥ 3, else last year's PPR
```

No market → zero w_m and renormalize. Situation multipliers (on top of OL / vacated / SOS / depth): cliff ×0.90, breakout ×1.06, team change ×0.97, role expand ×1.04, chronic injury ×0.96. Injury-bounce is still a tree feature; it is *not* a situation multiplier or a steal confirmer (it was anti-predictive).

6.3 Depth and vacated usage

Depth snapshot: historical week-1 REG (or last snapshot before Sept 8); 2026 = latest chart. Rank 1/2/3 weights 0.52 / 0.28 / 0.13. Vacated team share is split onto *people* by prior share + depth (rookies depth-only). player_vacated_boost = leftover targets for WR/TE, leftover carries for RB.

6.4 Red zone and TD luck (this is the fade “TD luck (over)”)

From PBP, dropping knees, spikes, penalties, 2-pt. Inside-5 carry: yardline_100 ≤ 5. Inside-10 target: ≤ 10. End-zone target: air_yards ≥ yardline_100 – 0.5 and yardline_100 ≤ 20.

```
HV-RZ = 1.4·carries_in5 + 1.0·targets_in10 + 1.6·ez_targets
rec_TD_luck = rec TDs - (0.20·targets_in10 + 0.28·ez_targets)
rush_TD_luck = rush TDs - 0.42·carries_in5
td_luck = rec_TD_luck + rush_TD_luck
```

Positive = scored more TDs than the role implied (binge). Negative = under-scored relative to looks. Fade confirmer: td_luck ≥ 2.0. Steal confirmer: td_luck ≤ -1. TD luck alone never paints a name; it only confirms a gap that is already ≥5 ranks and ≥12 PPR.

6.5 Other on/off flags (computed even if unused as steal/fade)

Flag	Definition	Steal/fade?
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workload_cliff	(carries ≥ 240 and age ≥ 26.5) or (carries ≥ 280 and age ≥ 26)	Fade confirmer
chronic_injury	≥ 8 missed-week flags in a 3-year rollup	Fade confirmer
overproduction	$z(\text{PPG}) - 0.5 \cdot (z(\text{snap}) + z(\text{target share})); \geq 0.55$	Fade confirmer
eff_index	sum of within-pos z of YPC, YPR, catch rate; ≥ 1.4	Fade confirmer
role_expand	listed starter, snap $< 55\%$, years_exp ≥ 1	Steal confirmer
sophomore_leap	years_exp = 1 and last PPR in [70, 200]	Steal confirmer
pass_catch_rb	RB target share ≥ 0.08	Steal confirmer
new_starter_vacated	new starter $\times$ vacated boost ≥ 6	Steal confirmer
injury_bounce	games_pct < 0.70 , age 23–31, last PPR/G ≥ 8	Tree only (stripped from flags; anti-predictive)
breakout_window	RB yr 1 / WR 1–2 / TE 2–3	Tree + situation $\times 1.06$; not a steal confirmer

6.6 Physical prior (young players)

```

speed_score = wt · 200 / forty4
burst       = vertical + broad_jump/12
draft_cap   = max(0.15, 1.15 - log(pick)/log(250))

```

Years_exp < 3 : pos_base · draft_cap · athletic (QB 220, RB 160, WR 140, TE 110). Veterans fall back to last year's PPR for the physical slot.

6.7 Added in the Aug 2026 enhancement pass

- **H2 deltas:** week ≥ 10 minus weeks 1–9 mean carries and PPR (lagged). Target-share H2 delta was ~ 0 leftover signal — skipped.
- **NGS:** intended air yards (QB), RYOE/att (RB), air-share (WR). Separation was ~ 0 — skipped.
- **QB change:** primary passer (PBP attempts; 2026 from depth QB1). wr_new_qb for WR/TE; new_starter_qb if this player is the new QB1. Head-coach change new_hc is kept in the trees.
- **ECR vs FFC ADP.**
- **new_oc:** Wikipedia OC names; PFR blocked. Partial history; 2026 current list is almost complete.

Luck z-score pieces (rec_td_luck, rush_td_luck, YPC/YPR/catch z) were dropped from the tree list; td_luck, overproduction, eff_index remain.

7. How the trees are actually weighted

Four HistGradientBoosting regressors, sklearn, random_state=7:

Tree	Target	Depth / iter / leaf
points	actual PPR	depth 3, 280 iter, min leaf 22, lr 0.05, L2 0.55
global residual	actual – market_fp	same, 220 iter; output clipped ± 60
position residual	same, per pos if $n \geq 120$	180 iter, leaf 18; mixed 55% local + 45% global
rank lift	ADP pos rank – actual pos rank	depth 2, 200 iter; clipped ± 18

Final mix when a market exists:

```
 $\hat{y} = \text{base} + \lambda_{\text{tenure}} \cdot \text{residual} + \text{clip}(k \cdot \text{lift}, -22, 22)$   
 $\lambda_{\text{tenure}} = \lambda \cdot 1.15 \text{ rookies, } \cdot 1.05 \text{ years } 1-2, \cdot 0.85 \text{ years } \geq 7, \text{ else } \lambda$   
base = VFP if vfp_markets  $\geq 1$  else ADP-implied points (else blend, else last PPR, else 80)
```

No market: $0.6 \cdot \text{points_tree} + 0.4 \cdot \text{blend}$.

Rookie gravity toward prior: extra 40% / 18% / 5% blend for years 0 / 1 / ≥ 2 , but if they have a real VFP that extra is capped at 8% (don't drag a posted rookie 40% off the books toward a combine prior).

λ and k are not hand-set. Grid $\lambda \in \{0.25, 0.40, 0.55, 0.75, 1.00\}$, $k \in \{0, 1.5\}$. Fit trees on seasons before the last training year; pick the pair that maximizes Spearman on ADP-drafted players in that last year. *Maximize ρ , not steal count.* Last 2026 fit used $k=0$ (rank-lift tree off). λ in backtests ranged 0.25 (2021) to 1.0 (2024).

Casual angle: turning the residual up until more names go green is how you lose to ADP. 2024's tuner chose $\lambda=1.0$ and still only tied ADP. The usable edge is disagreement, not "trust the trees more."

8. Steal / fade (the green and red pills)

Computed on *positional* ranks among ADP-drafted players, not overall VORP pick. Hover "i" lists up to three confirming phrases.

Steal if all of:

1. $\text{ADP} \in [18, 132]$ (skip 1.01s and dart throws)

2. model position rank $\leq$ ADP position rank $- 5$
3. $\hat{y} - m \geq 12$ PPR
4. at least one of: role expand, sophomore leap, pass_catch_rb ≥ 0.08 , td_luck ≤ -1 , new_starter_vacated ≥ 6

Fade if (2)–(3) flip sign and at least one of: workload cliff, chronic injury, td_luck ≥ 2.0 , overproduction ≥ 0.55 , eff_index ≥ 1.4 .

Early-round fades (ADP < 18): need fade_why and either ≥ 10 PPR below m, or a workload cliff with edge $\leq +5$. You do not fade a 1.01 because a tree twitched.

Walk-forward flag hit (stricter ≥ 4 actual rank gain metric in code): 10 steals, 60% hit, mean $+9.5$ ranks; 47 fades, 70% hit. Re-ranking inside the ADP pool, steal flags were closer to 80% beating ADP rank. Either way: small n on steals; the bigger published edge is *any* ≥ 5 -rank cheap call, not only painted names.

Cheap-call (not flag) walk-forward, ADP pool: ≥ 5 ranks cheaper beat ADP rank **76%** (68–93% every year). ≥ 8 cheaper: **81%**, mean $+12.5$ ranks. Mid-round ADP 36–96: historically model-cheap names scored more actual PPR than model-expensive names at the same cost (old figure ~ 177 vs 154).

9. How the hosted big board is ordered

This is the thing that broke when QBs flooded round 1.

- **Positional tabs:** sort by projected PPR (or rest-of-season PPR in season). QB1 = most points at QB. Correct.
- **Full board:** sort by **VORP** = $\hat{y} -$ replacement starter in a 12-team 1QB league. Roster is **2 RB, 2 WR, 1 FLEX** (plus 1 QB, 1 TE). Replacement ranks: **QB12, RB30, WR30, TE12** (12 flex spots split evenly: 24 starters + 6 flex at each of RB and WR). Same-position VORP order = points order. Across positions, elite RBs jump QBs because QB12 still scores ~ 270 PPR and RB30 does not. (An earlier 3-WR / WR36 version overstated how many WRs you have to start.)
- **vs ADP on the full board:** overall ADP rank $-$ overall pick (CMC pick 2 vs ADP $\sim 7 \rightarrow +5$). On positional tabs: ADP position rank $-$ model position rank (CMC RB2 vs ADP RB3 $\rightarrow +1$).
- **Steal/fade pills** still use the positional $\pm 5 / \pm 12$ rule + confirmer, not the overall vs number. Josh Allen can be $+20$ overall vs ADP and still be “fair” if he is QB1 vs ADP QB1.

Round headers: pick 1–12 = round 1, etc. 12-team snake. Early/mid/late within a round is picks 1–4 / 5–8 / 9–12 of that round.

10. Live overlay (does not retrain)

```
ros_fp = model_fp
    × (games left / scheduled)
    × (1 + 0.08 · SOS_remaining) / (1 + 0.08 · SOS_full)
    × indoor remaining/full for WR/QB
```

SOS is last year's opponent defensive EPA on this year's leftover opponents. After ~6 weeks of league games, blend in current-year YTD defense. Live team prefers Sleeper. Designations: IR / PUP / Q / O / D / SUS. Steal/fade labels stay frozen when names move. Daily Action: cron 0 14 * * * runs refresh_board.py and commits docs/index.html if it changed.

Toggles on the page: Full/Half PPR, Full/Positional, Start/Exit Draft (browser localStorage only). There is no separate midseason mode.

11. Walk-forward snapshot (last rebuild)

Year	n ADP	λ	k	Model ρ	ADP ρ	Δ	MAE PPR
2021	172	0.25	0	0.608	0.613	−0.005	56.6
2022	142	0.40	0	0.604	0.581	+0.023	53.3
2023	172	0.75	0	0.586	0.537	+0.049	57.1
2024	175	1.00	0	0.579	0.578	+0.001	59.8
2025	200	0.40	0	0.628	0.603	+0.026	54.7

Mean Spearman lift vs ADP $\approx$ +0.019. MAE 53–60 PPR is not a bug. Fair comparison is ADP-drafted players only.

Figures already generated (you can grab later): docs/figures/01–10 and copies in wordpress_figures/. Spearman, rank vs actual, lift heatmap, points vs actual, rank lift, flag hit rates, disagreement calibration, mid-round same cost, top-5 roster edge, cheap calls by year.

12. Example you already asked about (CMC vs Bijan)

Overall pick is VORP; both RBs share the same replacement line, so this is just 292 vs 286 PPR (~6 points). Vegas and ADP have Bijan first (VFP 308 vs 287; ADP 1.7 vs 6.5). The prior/blend also has Bijan higher (342 vs 321). The residual trees flip them because CMC's 417 PPR / 311

carries / 102 catches last year outweigh age-30 + workload cliff enough to land near his VFP, while Bijan is faded vs a 308 book / 1.01 ADP after `td_luck` 5.7 and `eff_index` 2.6. Neither is a steal/fade. Safe casual line: “six points, coin flip, the model is a little more scared of paying 1.01 for a TD binge than of a 30-year-old who just put up 417.”

13. Suggested article bones (you write the voice)

1. What the board is / isn't (ranking vs ADP, not a betting sheet).
2. The headline is disagreement, not “we beat ADP by 0.019 Spearman.”
3. Vegas does volume; we do aging, vacated *people*, TD luck vs role, cliffs.
4. How a name turns green/red (the three gates + one football reason). Mention TD luck as leftover vs red-zone role, threshold 2.0 for fades.
5. Why the first round is RBs/WRs (VORP / 2 RB, 2 WR, 1 FLEX — replacement QB still scores a ton).
6. Walk-forward: cheap calls hit most years; 2021/2024 were basically ties on full-board Spearman.
7. Half PPR is a tax, live injuries move the same table, not a new model.

14. File map if you need to check a fact

Path	What
<code>src/ffmodel/config.py</code>	scoring, peaks, tenure weights, steal thresholds, VORP replacement ranks
<code>src/ffmodel/features/assemble.py</code>	panel join (depth, H2, NGS, ECR, QB, OC)
<code>src/ffmodel/features/context.py</code>	VFP, age, vacated, situation, blend, TD luck, shrink
<code>src/ffmodel/features/edges.py</code>	depth snapshot, H2, NGS, QB, ECR, Wikipedia OC
<code>src/ffmodel/features/remaining.py</code>	<code>ros_fp</code> overlay
<code>src/ffmodel/model.py</code>	trees, λ/k , steal/fade, VORP attach
<code>src/ffmodel/publish.py</code>	board sort, vs ADP, hover why, HTML
<code>src/ffmodel/ingest/nflverse.py</code>	download paths
<code>src/ffmodel/ingest/markets.py</code>	FFC ADP, props CSV
<code>src/ffmodel/ingest/live_status.py</code>	Sleeper / nflverse badges

data/processed/rankings_2026.csv	current $\hat{y}$ (gitignored)
data/processed/backtest_by_season.csv	walk-forward table

15. Full tree feature list (FEATURE_COLS)

Trees see whatever of these exist on the panel, plus encoded `position_code` and `tenure_code`.
“Lag” means last season.

years_exp, age, age_alpha, games_lag, games_pct_lag, availability, ppr_lag, ppr_ppg_lag, attempts_lag, passing_yards_lag, passing_tds_lag, passing_epa_lag, passing_cpoe_lag, pacr_lag, carries_lag, rushing_yards_lag, rushing_tds_lag, rushing_epa_lag, rushing_20_lag, targets_lag, receptions_lag, receiving_yards_lag, receiving_tds_lag, receiving_epa_lag, target_share_lag, air_yards_share_lag, wopr_lag, receiving_20_lag, receiving_40_lag, explosive_rate, chunk_rating, chunk_index, hv_rz, hv_rz_per_touch, rz_index, inside5_carries, inside10_targets, ez_targets, player_vacated_targets, player_vacated_carries, player_vacated_boost, td_luck, catch_rate_lag, ypc_lag, ypr_lag, carry_load, carry_share_lag, pass_catch_rb, qb_rush_share, workload_cliff, breakout_window, sophomore_leap, team_change, new_hc_pass_catcher, pos_competition, starter, role_expand, injury_bounce, chronic_injury, overproduction, eff_index, young_capital, new_starter_vacated, usage_index, draft_capital, off_snap_pct, forty, speed_score, burst, bmi, draft_pick_num, log_draft_pick, undrafted, depth_rank, production_proxy, situation_proxy, physical_proxy, blend_proj, oline_index_shrunk, coaching_C_shrunk, proe_shrunk, pace_neutral_shrunk, sos_def_pass_epa_shrunk, sos_def_rush_epa_shrunk, sos_def_rec_epa_shrunk, pct_indoor_shrunk, avg_total_shrunk, avg_spread_for_shrunk, new_hc, carries_h2_delta, ppr_h2_delta, ngs_intended_air_yards, ngs_ryoe_att, ngs_air_share, qb_change, wr_new_qb, new_starter_qb, ecr, ecr_minus_adp, new_oc.

Not in the trees: rec_td_luck, rush_td_luck, eff_ypc_z, eff_ypr_z, eff_catch_z (dropped as redundant with td_luck / overproduction / eff_index). ADP and VFP are the market m , not tree columns (except via blend_proj / ecr).

Sources in one line for a footer: nflverse (CC BY 4.0); Fantasy Football Calculator ADP; publicly posted season-long totals; FantasyPros ECR via DynastyProcess; Wikipedia OC names. Not betting advice.